KS BHAVANA

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SUMMARY

Computer Science (Data Science) undergraduate with knowledge of Python, SQL, Machine Learning, and IoT. Hands-on experience through academic projects in AI, data analysis, and IoT, with an interest in solving real-world problems.

EDUCATION

B.Tech – Computer Science and Engineering (Data Science) 2023-2027
New Horizon College of Engineering, Bangalore | CGPA: 9.28

Class XII, Board of Intermediate Education, AP 2021-2023
Narayana Junior College, Tirupati | 95.5%

Class X, Board of Secondary Education, AP 2020-2021
Montfort Centenary School, Kuppam | 94.1%

TECHNICAL SKILLS

Programming: Python, Java, SQL
Data Science & ML: Pandas, NumPy, Matplotlib, Scikit-learn, XGBoost
Web Technologies: HTML, CSS, Flask
Industry 4.0: Odoo, ThingSpeak API, Browser Robotics, 3D Modelling
Core Concepts: Machine Learning, Federated Learning, Explainable AI
Tools: Jupyter Notebook, Tableau

PROJECTS

Privacy Preserving Federated Intrusion Detection System For Real Time IOT Networks
Python | XGBoost | Random Forest | ANN/MLP | Federated Learning | SHAP

- Developed a privacy-preserving intrusion detection system for IoT networks using Federated Learning.
- Implemented and compared XGBoost, Random Forest, and ANN/MLP models to classify network traffic and detect cyberattacks.
- Integrated SHAP-based Explainable AI to identify important features contributing to attack predictions.

DERMA-AI – Smart Skin Health & Lifestyle Guidance System
Python | Flask | Computer Vision | LLaMA | HTML | CSS

- Developed an AI-based application that analyzes skin images and provides personalized skincare and lifestyle recommendations.
- Integrated computer vision and LLaMA to analyze uploaded images and generate user-specific insights.
- Built a Flask-based web application with user registration, image upload, analysis, and result reporting.

AI-Powered Company-Specific Interview Guide
Python | Flask | LLaMA 3.2 | HTML | CSS

- Developed an AI-powered web application that generates company-specific interview preparation content.
- Integrated LLaMA 3.2 with a Flask backend to generate interview questions and preparation guidance.
- Designed a simple web interface for users to enter company and role details and receive customized preparation.

CERTIFICATIONS

- Internet of Things (IoT) Certification | Infosys Springboard
- SAP Analytics Cloud – Beginner Level | SAP Learning | April 2026
- Fundamentals of Operating Systems-NPTEL

ACADEMIC EXPERIENCE

Project Experience

Industry 4.0 – IoT-Based Temperature & Humidity Monitoring System

- Developed an IoT-based system to monitor real-time temperature and humidity using a DHT sensor.
- Implemented MQTT for real-time communication and transmission of sensor data.
- Integrated ThingSpeak for cloud-based data storage, monitoring, and visualization.
- Displayed real-time environmental data through an IoT monitoring dashboard.

Industry 4.0 Technical Experience

- **IoT & Sensors:** Worked with DHT sensors for real-time environmental data collection and monitoring.
- **MQTT & ThingSpeak:** Implemented MQTT-based sensor data communication and ThingSpeak for cloud-based monitoring and visualization.
- **Odoo ERP:** Applied ERP concepts for understanding business operations, workflows, and organizational processes.
- **Browser Robotics:** Applied browser automation concepts for automating web-based tasks and interactions.
- **Fusion 360:** Created 3D models and explored CAD-based design concepts.
- **CNC Manufacturing:** Studied computer-controlled machining and automated manufacturing processes.